BCS-076	Distributed Database	L T P 3 0 0	3 Credits

Pre-requisites: students should know about the database and distribution system and how the data can distribute.

Course Objectives:

CO1	Enhanced the knowledge in the area of Distributed Database system.
CO2	Comprehend the Distributed query processing
CO3	The subject explores the ideas of Transaction management and concurrency control.
CO4	Know the parallel database system architecture.
CO5	To learn about Distributed Deadlock Detection

Detailed Syllabus

MODULE-I

Transaction and schedules: Concurrent Execution of transaction, Conflict and ViewSerializability, Testing for Serializability, Concepts in Recoverable and Cascadeless schedules.
Lock based protocols, time stamp based protocols, Multiple Granularity and MultiversionTechniques, Enforcing serializablity by Locks, Locking system with multiple lock modes, architecture for Locking scheduler.

MODULE-II

Distributed Transactions Management: Data Distribution, Fragmentation and ReplicationTechniques, Distributed Commit, Distributed Locking schemes, Long duration transactions,Moss Concurrency protocol.
Issues of Recovery and atomicity in Distributed Databases, Traditional recovery techniques, Logbased recovery, Recovery with Concurrent Transactions, Recovery in Message passing systems, Checkpoints, Algorithms for recovery line, Concepts in Orphan and Inconsistent Messages.

MODULE-III

Distributed Query Processing: Multiway Joins, Semi joins, Cost based query optimization for distributed database, Updating replicated data, protocols for Distributed Deadlock Detection, Eager and Lazy Replication Techniques.

Course Outcomes: After the completion of the course the student will be able to:

CO1	Aware of fundamentals of Distributed Database systems.
CO2	Use the different techniques of Distributed query processing.
CO3	Set the rules over management of transaction and concurrency control.
CO4	Familiar with parallel database system architecture.
CO5	Understand Distributed Deadlock Detection

Text Books:

1. Silberschatz, orth and Sudershan, Database System Concept', McGraw Hill
2. Ramakrishna and Gehrke,' Database Management System, McGraw Hill

References:

1.Garcia-Molina, Ullman,Widom,' Database System Implementation' Pearson Education
2. Ceei and Pelagatti,'Distributed Database', TMH
3. Singhal and Shivratri, 'Advance Concepts in Operating Systems' MC Graw Hill